

Nazariy masala 1

RELATIVISTIK ZARRA

Maxsus nisbiylik nazariyasida tinch massasi m_0 bo'lgan erkin zarrachaning energiyasi E va impulsi P orasidagi munosabat:

$$E = \sqrt{p^2 c^2 + m_0^2 c^4} = mc^2.$$

Bunday zarracha konservativ kuch ta'sirida bo'lganda, $\sqrt{p^2 c^2 + m_0^2 c^4}$ va potensial energiya yig'indisi (to'liq energiya) saqlanadi. Agar zarrachaning energiyasi juda katta bo'lsa, tinch energiyani e'tiborsiz qoldirish mumkin (bunday zarracha *ultrarelativistik* zarracha deb ataladi).

1. O'zgarmas kattalikdagi f markazga tortuvchi kuch ta'siridagi bir o'lchovli harakatni qaraylik. Zarracha dastlab kuch markazida bo'lib, $t = 0$ da boshlang'ich impulsi p_0 ga ega. Zarrachaning harakatini tasvirlang: $x(t)$ va $p(x)$ grafiklarini harakatning kamida bir davri uchun alohida chizing. Berilgan p_0 va f parametrlari orqali "burilish nuqtalari" koordinatalarini aniqlang. (p, x) diagrammasida harakat yo'nalishini o'qlar bilan ko'rsating. Zarracha ultrarelativistik bo'lmagan qisqa vaqt oralig'ini e'tiborsiz qoldiring.

Javob varag'i 1 dan foydalaning.

2. Mezon – bu ikki kvarkdan tashkil topgan zarrachadir. Mezonning tinch massasi M ikki kvark tizimining umumiy energiyasini c^2 ga bo'linganiga teng.

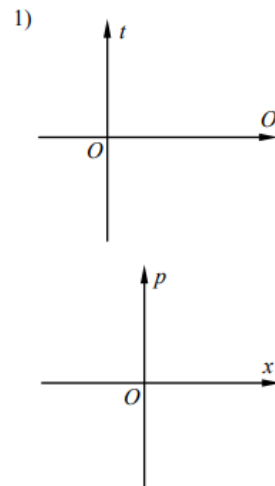
Tinch turgan mezonning bir o'lchovli modelini qaraylik. Ikki kvark x o'qi bo'ylab harakatlanadi va bir-biriga o'zgarmas kattalikdagi f kuchi bilan tortishadi. Ular bir-biridan erkin o'ta oladi deb faraz qilinadi. Kvarklarning yuqori energiyali harakatini tahlil qilishda ularning tinch massasini e'tiborsiz qoldirish mumkin. $t = 0$ da ikkala kvark $x = 0$ da joylashgan. Ikki kvarkning harakatini (x, t) diagrammasi va (p, x) diagrammasida grafik ravishda alohida ko'rsating. "Burilish nuqtalari" koordinatalarini M va f orqali aniqlang, (p, x) diagrammasida jarayon yo'nalishini ko'rsating va ikki kvark orasidagi maksimal masofani toping.

Javob varag'i 2 dan foydalaning.

3. 2-qismda ishlatilgan sanoq sistemasi S ga (laboratoriya sistemasi) nisbatan manfiy x yo'nalishida $v = 0.6c$ doimiy tezlik bilan harakatlanuvchi S' sistemani qaraylik. Ikkala sanoq sistemasining koordinatalari shunday tanlanganki, S' sistemada mezonning impulsi $p' = 0$ bo'lsin. S' sistemada ikki kvark orasidagi maksimal masofani toping. S' sistemada mezonning energiyasini hisoblang (massa $M = 140 \text{ MeV}/c^2$ deb oling).

Javob varag'i 3 dan foydalaning.

ANSWER SHEET 1



ANSWER SHEET 2

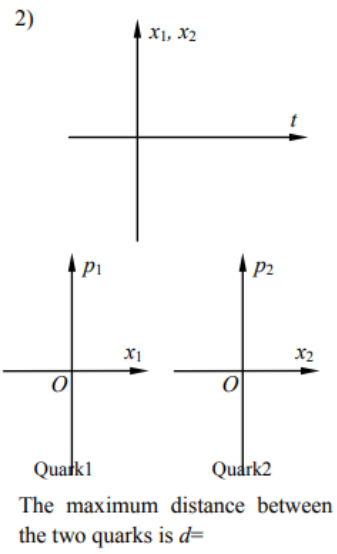
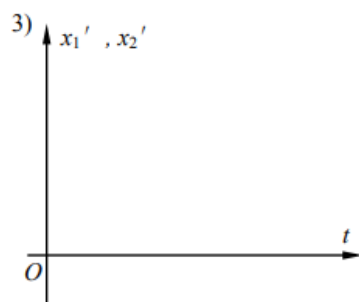


Figure 1: 1-rasm (sahifa 1)

ANSWER SHEET 3



The maximum distance between the two quarks observed in S' frame is $d' =$

3)

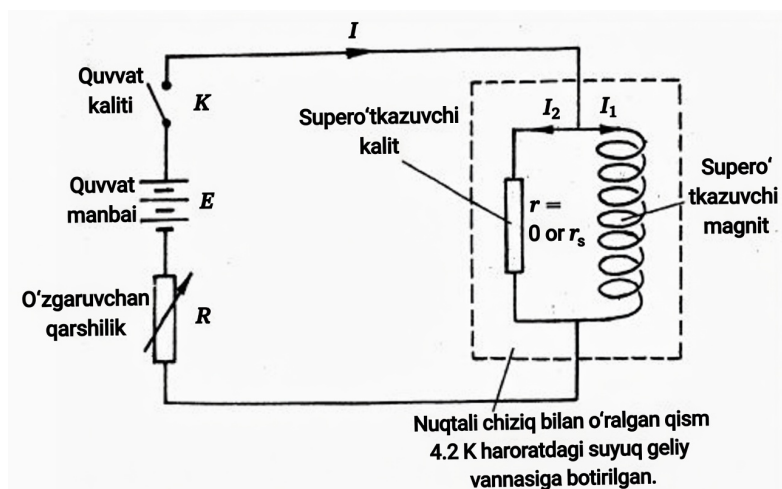
$$x'_1, x'_2$$

Ikki kvark orasidagi S' sistemasida kuzatilgan maksimal masofa $d' = \dots$

Nazariy masala 2

SUPERO'QKAZGICH MAGNITLAR

Supero'tkazgich magnitlar laboratoriyalarda keng qo'llaniladi. Eng keng tarqalgan shakli – supero'tkazgich simdan yasalgan solenoid. Supero'tkazgich magnitning ajoyib xususiyati shundaki, u Joule issiqligi tufayli energiya yo'qotilmasdan yuqori magnit maydonlarini hosil qiladi, chunki magnit geliy suyuqligiga (4.2 K) botirilganda simning elektr qarshiligi nolga aylanadi. Odatda, magnit 1-rasmda ko'rsatilganidek, maxsus ishlab chiqilgan supero'tkazgich kalit bilan ta'minlanadi. Kalitning qarshiligi r boshqarilishi mumkin: supero'tkazgich holatida $r = 0$, normal holatida esa $r = r_n$. Kalit magnit va supero'tkazgich kalit orqali aylanib yuruvchi tok bilan "persistent" rejimda ishlaganda, tashqi manba uzilgan holda magnit maydon uzoq vaqt saqlanishi mumkin.



1-rasmda supero'tkazgich kalitning tafsilotlari berilmagan. Odatda bu qisqa uzunlikdagi supero'tkazgich sim bo'lib, unga isitgich sim o'ralgan va suyuq geliy vannasidan issiqlik izolyatsiyasi ta'minlangan. Qizdirilganda, supero'tkazgich simning harorati oshadi va u qarshilikka ega normal holatga o'tadi. r_n ning odatdagi qiymati bir necha ohm. Bu yerda biz uni 5Ω deb qabul qilamiz. Supero'tkazgich magnitning induktivligi uning o'lchamiga bog'liq; 1-rasmdagi magnit uchun uni 10 H deb faraz qilaylik. Umumiy tok I R qarshilikni sozlash orqali o'zgartirilishi mumkin.

Ushbu masala faqat grafiklar asosida baholanadi!

O'qlar I , I_1 va I_2 ning musbat yo'nalishini bildiradi.

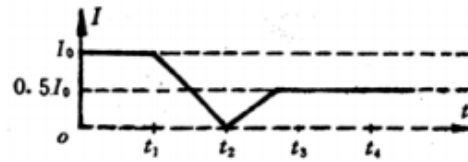
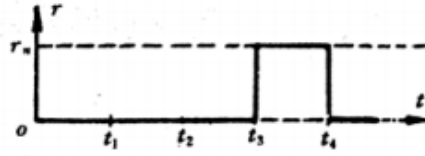
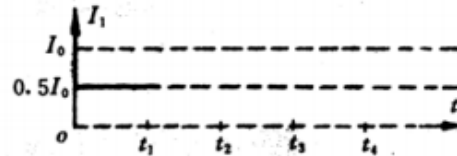


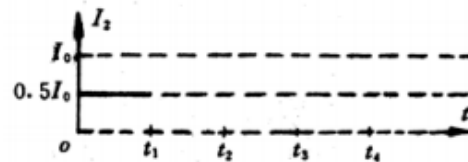
Fig.2a



2b



2c



2d

Figure 2: 1-rasm. Supero'tkazgich kalit.

1. Agar umumiy tok I va supero'tkazgich kalitning qarshiligi r vaqt bo'yicha mos ravishda 2a va 2b-rasmlarda ko'rsatilganidek o'zgarsa va magnit va kalit orqali o'tadigan I_1 va I_2 toklari boshida teng deb faraz qilinsa (2c va 2d-rasmlar), ular t_1 dan t_4 gacha qanday o'zgaradi? Javobingizni 2c va 2d-rasmlarga chizing.
2. Faraz qilaylik, quvvat kaliti K $t = 0$ da yoqiladi, bunda $r = 0$, $I_1 = 0$ va $R = 7.5 \Omega$, umumiy tok $I = 0.5$ A. K yopiq holda, supero'tkazgich kalitning qarshiligi r 3b-rasmda ko'rsatilganidek o'zgartiriladi. I , I_1 va I_2 ning vaqtga bog'liqliklarini mos ravishda 3a, 3c va 3d-rasmlarga chizing.
3. Supero'tkazgich kalit normal holatda bo'lganida undan faqat kichik toklar (0.5 A dan kam) o'tishi mumkin, katta toklarda kalit yonib ketadi. Supero'tkazgich magnit 4-rasmda ko'rsatilganidek, $t = 0$ dan $t = 3$ min gacha persistent rejimda ishlayapti, ya'ni $I = 0$, $I_1 = I_0$ (masalan 20 A), $I_2 = I_1$. Agar tajribani magnit orqali tokni nolga kamaytirish orqali to'xtatmoqchi bo'lsak, buni qanday qilish kerak? Bu bir necha amalda bajarilishi kerak. I , r , I_1 va I_2 ning mos o'zgarishlarini 4-rasmga chizing.
4. Faraz qilaylik, magnit 20 A persistent tok bilan persistent rejimda ishlayapti ($t = 0$ dan $t = 3$ min gacha, 5-rasmga qarang). Uni 30 A li persistent rejimga qanday o'zgartirasiz? Javobingizni 5-rasmga chizing.

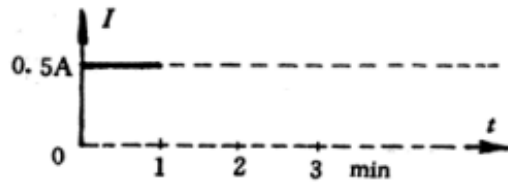
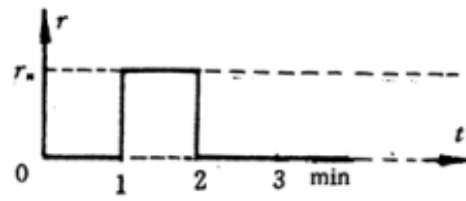
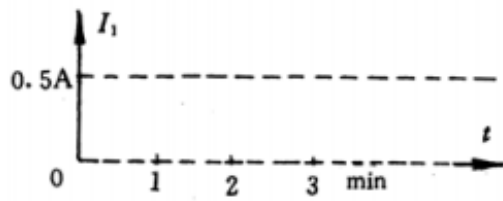


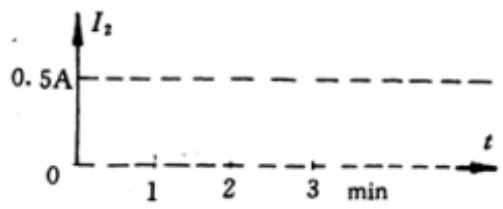
Fig. 3a



3b



3c



3d

Figure 3: 2a-2d rasmlar

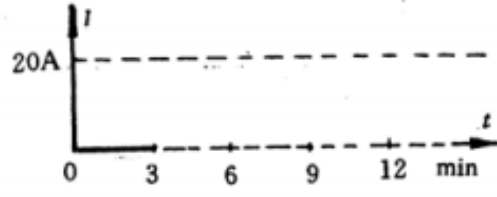
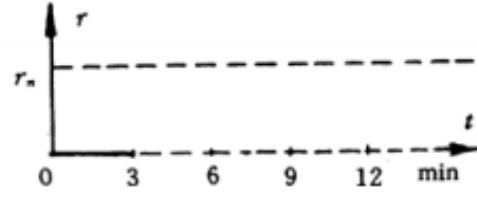
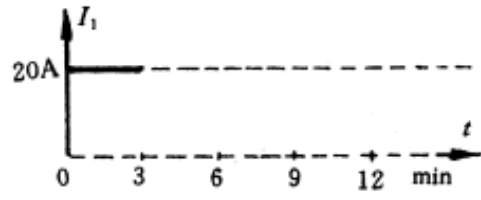


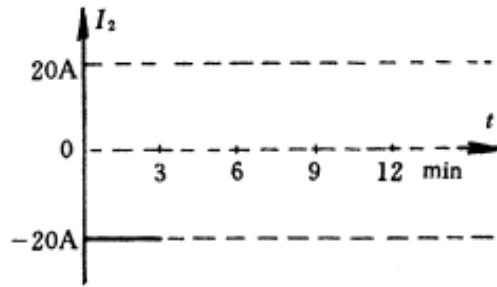
Fig. 4a



4b



4c



4d

Figure 4: 3a-3d va 4a-4d rasmlar

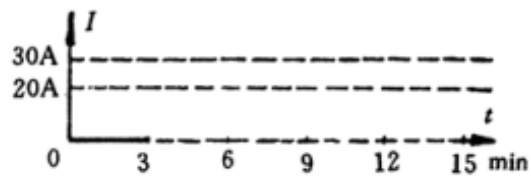
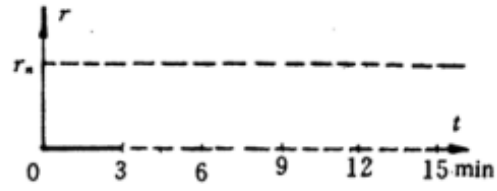
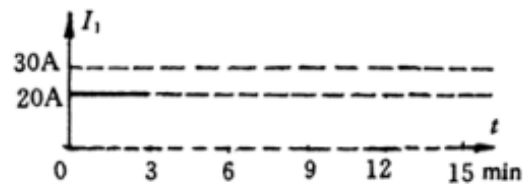


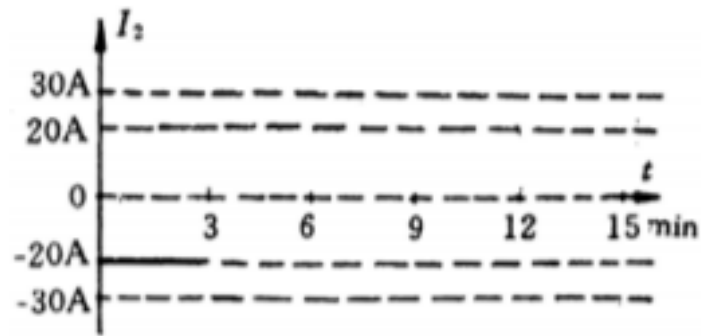
Fig. 5a



5b



5c



5d

Figure 5: 5a-5d rasmlar

Nazariy masala 3

DISKLARNING SIRT ISHQALANISHI BILAN TO'QNASHUVI

Bir jinsli A diski massasi m , radiusi R_A bo'lib, silliq gorizontal $x - y$ tekisligida x yo'nalishida V tezlik bilan ilgarilanma harakat qiladi (keyingi sahifadagi rasmga qarang). Disk markazi x o'qidan b masofada joylashgan. U koordinata sistemasining boshida markazi joylashgan tinch turgan bir jinsli B diski bilan to'qnashadi. B diski bir xil massa va qalinlikka ega, lekin radiusi R_B . To'qnashuvdan keyin disklarning tutashish nuqtasidagi tezliklari, markazlarni tutashtiruvchi chiziqqa perpendikulyar yo'nalishda teng bo'ladi deb faraz qilinadi. Shuningdek, disklarning markazlarni tutashtiruvchi chiziq bo'ylab nisbiy tezliklarining kattaliklari to'qnashuvdan oldin va keyin bir xil deb faraz qilinadi.

1. Bunday to'qnashuv uchun ikki diskning to'qnashuvdan keyingi tezliklarining X va Y komponentlarini, ya'ni V'_{AX} , V'_{AY} , V'_{BX} va V'_{BY} ni m , R_A , R_B , V va b orqali aniqlang.
2. To'qnashuvdan keyin A diskning kinetik energiyasi E'_A va B diskning kinetik energiyasi E'_B ni m , R_A , R_B , V va b orqali aniqlang.

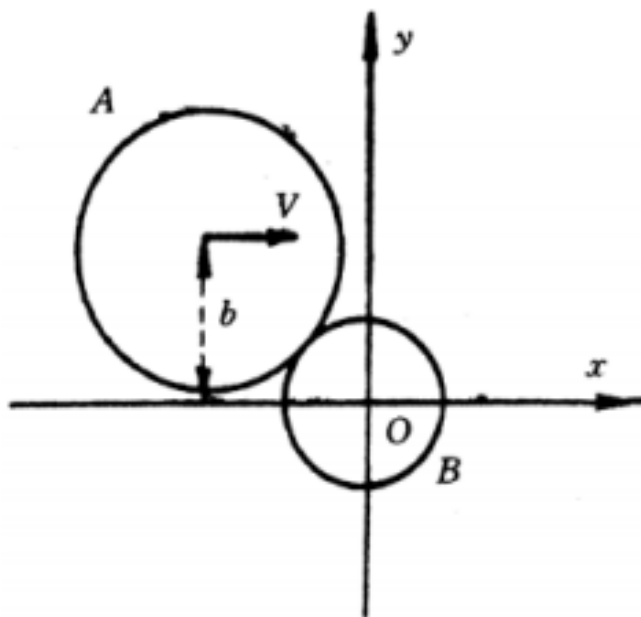


Figure 6: To'qnashuv sxemasi

XXV XALQARO FIZIKA OLIMPIADASI
PEKIN, XITOIY XALQ RESPUBLIKASI
AMALIY MUSOBAQA

1994-yil 15-iyul

Mavjud vaqt: 2.5 soat

AVVAL BUNI O'QING!

KO'RSATMALAR:

1. Faqat berilgan sharikli qalamdan foydalaning.
2. Grafiklaringiz masalaga biriktirilgan javob varaqlariga chizilishi kerak.
3. Yechimingizni faqat qog'ozning belgilangan tomoniga yozing.
4. Qoralama qog'ozlar sonli hisoblar va qoralama chizmalar uchun berilgan.
5. Har bir varaqning tepasiga yozing:
 - masala raqami,
 - har bir masala bo'yicha hisobotingizning varaq raqami,
 - masalaga oid hisobotingizdagi varaqlarning umumiy soni,
 - ismingiz va kod raqamingiz.

AMALIY MASALA 1

Shaffof dielektrik sirtining yorug'lik qaytarish koeffitsientini aniqlash

Eksperimental jihozlar

1. He-Ne lazer (1.5 mV). Ushbu lazerning nuri chiziqli qutblangan emas.
2. Darajali shkala diski bo'lgan ikkita qutblagich (P_1 , P_2) (1-rasm). Ulardan biri (P_1) lazer chiqish oynasi oldiga qutblagich sifatida o'rnatilgan, ikkinchisi esa kerak bo'lganda chizma taxtasiga qadama qog'oz qadamlari bilan mahkamlanishi mumkin.
3. Yorug'lik intensivligining ikkita detektori (D_1 , D_2) – foto element va mikroampermetrdan iborat (2-rasm).
4. Shisha nur ajratgich (B).
5. Qaytarish koeffitsienti va sindirish ko'rsatkichi aniqlanishi kerak bo'lgan shaffof dielektrik plastinka.
6. Koaksiyal aylanadigan qo'l bilan yarim doira shkalali plastinkaga o'rnatilgan namuna stoli (3-rasm).
7. Namunaviy stolni chizma taxtasiga mahkamlash va uning aylanish o'qi sifatida bir nechta qadama qog'oz qadamlar.
8. Lazer nurini gorizontal yo'nalishda sozlash va optik elementlarni moslashtirish uchun tirqish diafragma va ekran.
9. Optik elementlarni belgilangan joyga yopishtirish uchun lyute.
10. Yog'och chizma taxtasi.
11. Grafik qog'ozlar.

Eksperiment talablari

1. p -komponentaning qaytarish koeffitsientini tushish burchagiga bog'liq holda aniqlang (p -komponenta – elektr maydon vektorining tushish tekisligiga parallel bo'lgan tashkil etuvchisi).
 - (a) p -komponenta o'lchashida qutblagichning o'tkazish o'qini (A) darajali shkala diskidagi belgilangan chiziqning holati orqali aniqlang (o'tkazish o'qi – o'tgan yorug'likning elektr maydon vektori tebranish yo'nalishi).
 - (b) Yorug'lik intensivligi detektorlaridan birini tanlang va uning mikroampermetrini " $\times 5$ " diapazoniga qo'ying. Yorug'lik intensivligi va mikroampermetr ko'rsatkichi o'rtasidagi chiziqli bog'liqlikni tekshiring. Optik sxema diagrammasini chizing. O'lchangan ma'lumotlaringizni va hisoblangan natijalaringizni (hisoblash formulasi bilan birga) jadval shaklida ko'rsating. Chiziqli bog'liqlik egri chizig'ini chizing.
 - (c) p -komponentaning qaytarish koeffitsientini tushish burchagiga bog'liq holda aniqlang. Optik sxema diagrammasini chizing. O'lchangan ma'lumotlaringizni va hisoblangan qaytarish koeffitsientlarini (hisoblash formulasi bilan birga) jad-

val shaklida ko'rsating. Qaytarish koeffitsientini tushish burchagiga bog'liq holda chizing.

2. Namunaning sindirish ko'rsatkichini iloji boricha aniq aniqlang.

Tushuntirish va maslahatlar

1. Lazer nurlanishi, to'g'ridan-to'g'ri ko'zga tushishidan saqlaning.
2. Lazer nurining chiqish quvvati vaqt o'tishi bilan o'zgarib turishi mumkinligi sababli, tajriba davomida yorug'lik chiqishining tebranishini kuzatib borish va eksperimental natijalarni tuzatish kerak.
3. Lazer butun vaqt davomida yoniq bo'lishi kerak, hatto tajribangizni tugatib, imtihon zalini tark etganingizda ham lazer ishlashda davom etishi kerak.
4. θ_B tushish burchagida aks ettirilgan yorug'lik to'liq tekis qutblangan bo'ladi va $\tan \theta_B = n$ (sindirish ko'rsatkichi).

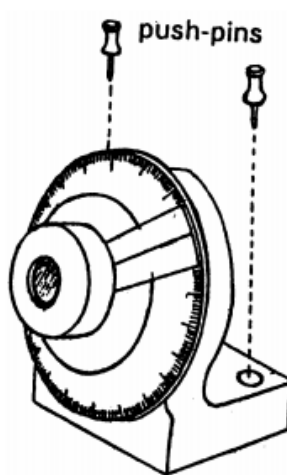


Fig. 1 polarizers with degree scale disk

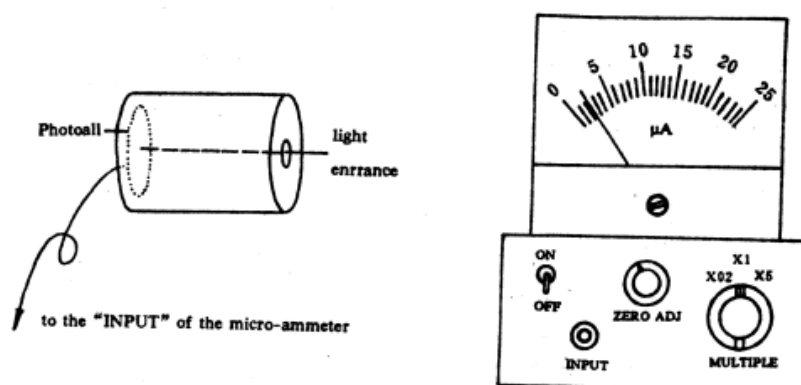


Fig. 2 Light intensity detector

Figure 7: 1-rasm: Darajali shkala diskli qutblagichlar, 2-rasm: Yorug'lik intensivligi detektori

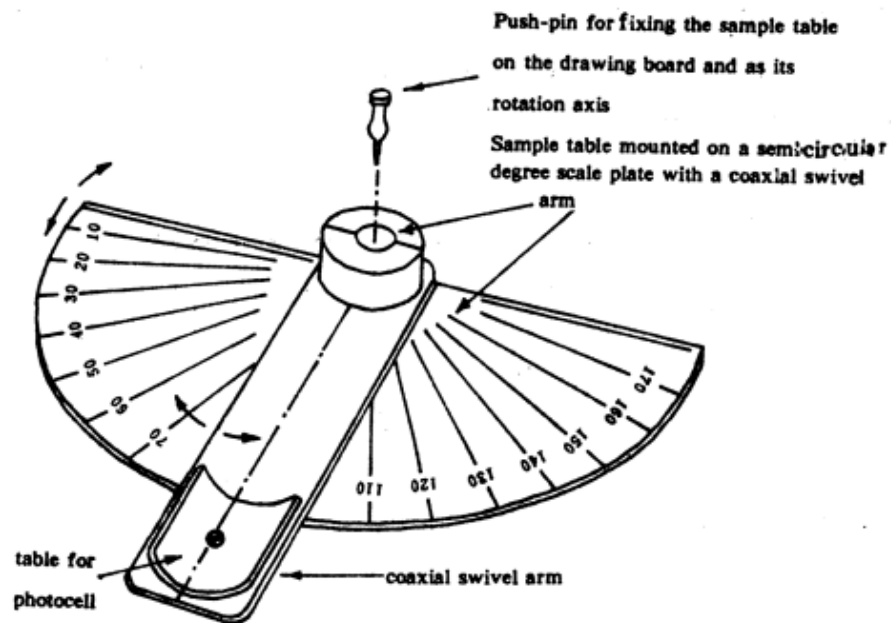


Fig.3 Sample table mounted on a semicircular degree scale plate

Figure 8: 3-rasm: Namunaviy stol va yarim doira shkalali plastinka

AMALIY MASALA 2

QORA QUTI

Ikki o'xshash terminalga ega qora quti berilgan. Qora quti ichida uchta ko'p bo'lmagan passiv elementlar mavjud. Terminallar orasidagi ekvivalent zanjir elementlarining qiymatlarini toping. Ushbu qutini ochishga ruxsat berilmaydi.

Eksperimental jihozlar

1. Har bir tugmachaning nomi va funksiyasini ko'rsatadigan panel tasviri bilan ikki kanalli osillograf.
2. Har bir tugmachaning nomi va funksiyasini ko'rsatadigan panel tasviri bilan audio chastota signal generatori.
3. 100 ohm ($<\pm 0.5\%$) qarshilik qutisi.
4. Bir nechta ulash simlari.
5. Koaksiyal kabellar uchun terminaldagi qora sim yerga ulangan.
6. Kerak bo'lganda foydalanish uchun log-log, yarim log va millimetr qog'ozlari berilgan.

Eslatma: "Signal generatori" va "osillograf"ning panel tasvirida ko'rsatilmagan tugmachalar to'g'ri holatlarga o'rnatilgan. Talaba ularga tegmasligi kerak.

Eksperimental talablar

1. Tajribangizdagi zanjir diagrammasini chizing.
2. O'lchangan ma'lumotlaringizni va hisoblangan natijalaringizni jadvallar shaklida ko'rsating. Olingan natijalar bilan eksperimental egri chiziqlarni taqdim etilgan koordinata diagrammalarida chizing (diagrammaning sarlavhasini, koordinata o'qlarining sarlavhalari va shkala birliklarini ko'rsating).
3. Qora qutining ekvivalent zanjirini va undagi elementlarning nomlari va qiymatlarini bering (hisoblash formulalarini yozing).

Ko'rsatmalar

1. Tajribangizni 100 Hz dan 50 kHz gacha chastota oralig'ida bajaring.
2. Signal generatorining chiqish kuchlanishi 1.0 V (peyk-to-peyk) dan kam bo'lishi kerak. "Out Attenuation" kalitini "20" dB holatiga qo'ying va uni o'zgartirmang.
3. Simlarni ulashda, elektr tarmog'idan 50 Hz interferensiyani minimallashtirish uchun ehtiyot bo'ling.

XD2 tipidagi chastota generatoridan foydalanish bo'yicha ko'rsatma

1. "Out Attenuation" ni "20" dB holatiga qo'ying va o'zgartirmang.
2. "Damping Switch" ni "Fast" holatiga qo'ying.
3. Signal generatorining voltmetr ko'rsatkichi nisbiy qiymatdir, haqiqiy chiqish qiymati emas.
4. Chastota ko'rsatkichlari xatosini e'tiborsiz qoldiring.

Eslatma: XD22 tipidagi audio chastota generatori uchun "Damping Switch" yo'q va "output" kaliti sinus " " holatiga qo'yilishi kerak.

SS-5702 tipidagi osillografdan foydalanish bo'yicha ko'rsatma

1. "V mode" kalitini "Dual" holatida saqlang.
2. "Volts/div" (qora) va "variable control" (qizil) vertikal kuchaytirgichning kuchaytirishini o'zgartiradi va "variable control" (qizil) to'liq soat yo'nalishida bo'lganda, qora sozlamalar kalibrlangan hisoblanadi.
3. "Times/div" (qora) gorizontaal supurish tezligini $0.5 \mu\text{s}/\text{div}$ dan $0.2 \text{ s}/\text{div}$ gacha o'zgartiradi va ular "variable control" (qizil) to'liq soat yo'nalishidagi CAL holatida bo'lganda kalibrlangan hisoblanadi.
4. "Triggering Source" supurish signalini tanlash uchun, "level" esa trigger signalining amplitudasini sozlash uchun ishlatiladi.
5. O'lchash aniqligi: $\pm 4\%$.

"Qarshilik qutisi" dan foydalanish bo'yicha ko'rsatma Qarshilik qutisining qarshiligi 100 ohm ga o'rnatilgan va uni o'zgartirmaslik kerak.

Nazariy masala 1 – Yechim

- 1) 1a. Kuch markazini x koordinatasining boshi va potensial energiyaning nol nuqtasi deb olib, zarrachaning potensial energiyasi

$$U(x) = f|x| \quad (1)$$

To'liq energiya

$$W = \sqrt{p^2 c^2 + m_0^2 c^4} + f|x|.$$

- 1b. Tinch energiyani e'tiborsiz qoldirib,

$$W = |p|c + f|x|, \quad (2)$$

W harakat davomida saqlanganligi sababli,

$$W = |p|c + f|x| = p_0 c, \quad (3)$$

x o'qini zarrachaning boshlang'ich impulsi yo'nalishida olamiz:

$$\begin{aligned} pc + fx &= p_0 c & x > 0, p > 0; \\ -pc + fx &= p_0 c & x > 0, p < 0; \\ pc - fx &= p_0 c & x < 0, p > 0; \\ -pc - fx &= p_0 c & x < 0, p < 0. \end{aligned} \quad (4)$$

Zarrachaning boshlang'ich nuqtadan maksimal masofasi L , $p = 0$ ga mos keladi:

$$L = p_0 c / f.$$

- 1c. (3) tenglama va Nyuton qonunidan

$$\frac{dp}{dt} = F = \begin{cases} -f, & x > 0; \\ f, & x < 0; \end{cases} \quad (5)$$

zarrachaning tezligini topamiz:

$$\left| \frac{dx}{dt} \right| = \left| \frac{c}{f} \frac{dp}{dt} \right| = c, \quad (6)$$

ya'ni juda yuqori energiyali zarracha $x = \pm L$ nuqtalariga juda yaqin hududdan tashqari har doim yorug'lik tezligida harakat qiladi. Zarrachaning markazdan $x = L$ nuqtasiga borish vaqti τ :

$$\tau = L/c = p_0/f.$$

Shunday qilib, zarracha $x = L$ va $x = -L$ oralig'ida c tezlik bilan oldinga-orqaga harakat qiladi, davri $4\tau = 4p_0/f$.

x va t orasidagi munosabat:

$$\begin{aligned} x &= ct, & 0 \leq t \leq \tau \\ x &= 2L - ct, & \tau \leq t \leq 2\tau, \\ x &= 2L - ct, & 2\tau \leq t \leq 3\tau, \\ x &= ct - 4L, & 3\tau \leq t \leq 4\tau. \end{aligned}$$

Kerakli javoblar 1-rasm va 2-rasmda berilgan.

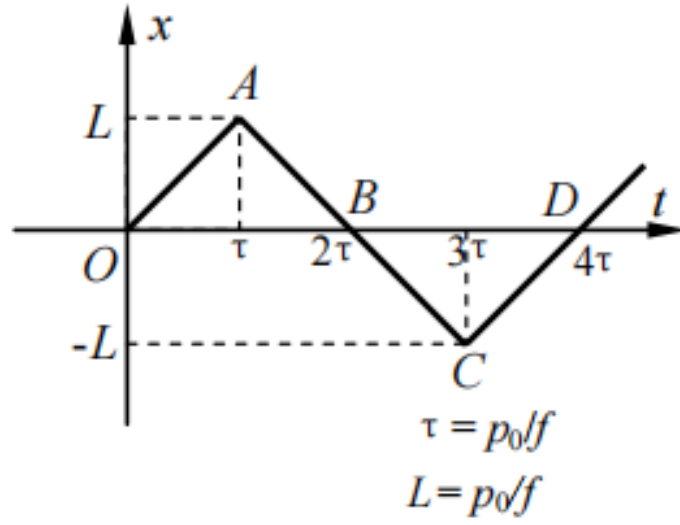


Fig. 1

Figure 9: 1-rasm: $x(t)$

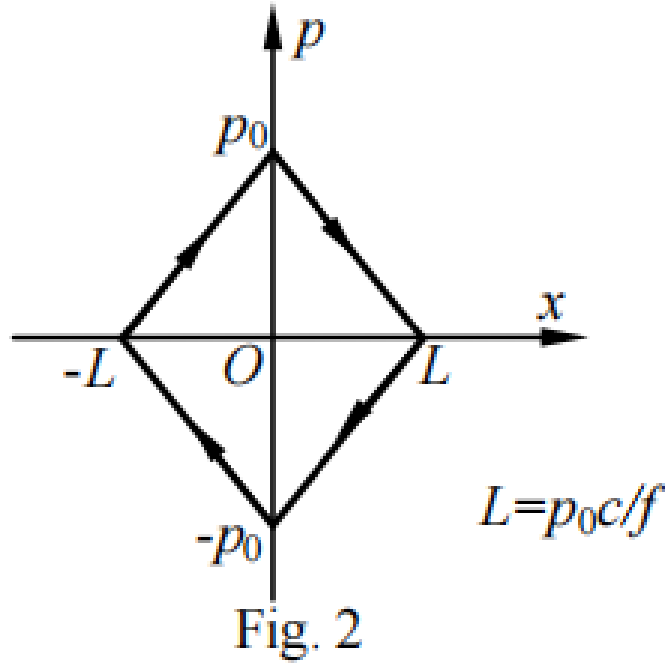


Fig. 2

Figure 10: 2-rasm: $p(x)$

2) Ikki kvark tizimining to'liq energiyasi

$$Mc^2 = |p_1|c + |p_2|c + f|x_1 - x_2|,$$

bu yerda x_1, x_2 – kvarklarning koordinatalari, p_1, p_2 – ularning impulsleri. Tinch turgan mezon uchun ikki kvarkning umumiy impulsi nolga teng va kvarklar simmetrik ravishda

qarama-qarshi yo'nalishda harakatlanadi:

$$p = p_1 + p_2 = 0, \quad p_1 = -p_2, \quad x_1 = -x_2.$$

p_0 ni $x = 0$ da 1-kvarkning impulsi deb belgilaymiz:

$$Mc^2 = 2p_0c \quad \text{yoki} \quad p_0 = Mc/2.$$

(8), (9), (10) dan 1-kvarkning energiyasining yarmi:

$$p_0c = |p_1|c + f|x_1|,$$

bu 1-qismdagi (3) tenglamaga o'xshaydi, boshlang'ich impulsi p_0 va $x = 0$ da joylashgan.

Shuning uchun 1-kvarkning harakati avvalgi kabi, massasi e'tiborga olinmaydigan

ultrarelativistik zarrachaniki kabi bo'ladi. 2-kvark esa unga nisbatan simmetrik harakatlanadi. Kvarklarning (x, t) diagrammasi 3-rasmda, (p, x) diagrammasi 4-rasmda ko'rsatilgan.

Burilish nuqtalari koordinatalari:

$$x_1 = \pm L, \quad x_2 = \mp L, \quad \text{bunda} \quad L = p_0c/f = Mc^2/(2f).$$

Ikki kvark orasidagi maksimal masofa:

$$d_{\max} = 2L = Mc^2/f.$$

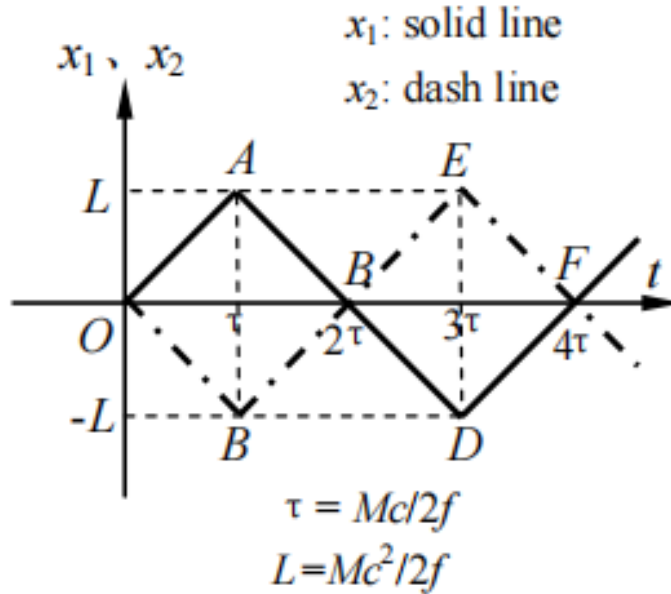


Fig. 3

Figure 11: 3-rasm: $x_1(t)$, $x_2(t)$

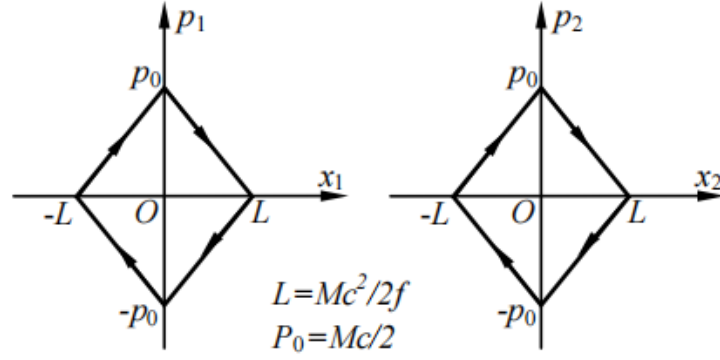


Fig. 4a

Quark1

Fig. 4b

Quark2

Figure 12: 4-rasm: $p(x)$

3)

S sanoq sistemasi laboratoriya sistemasiga nisbatan $V = 0.6c$ ozgarmas tezlik bilan x' oqi yonalishida harakatlanadi va ikkala sistemaning boshlanish nuqtalari boshida ($t = t' = 0$) ustma-ust tushadi. Bu ikki sistema orasidagi Lorents almashtirishlari quyidagicha:

$$x' = \gamma(x + \beta t), \quad t' = \gamma(t + \beta x/c),$$

bu yerda $\beta = V/c$, $\gamma = 1/\sqrt{1 - \beta^2}$. $V = 0.6c$ bolganda $\beta = 3/5$, $\gamma = 5/4$. Lorents almashtirishi chiziqli bolgani uchun (x, t) diagrammasidagi togri chiziq (x', t') diagrammasida ham togri chiziqqa aylanadi, shuning uchun biz faqat S' sistemasidagi burilish nuqtalarining koordinatalarini hisoblashimiz kerak.

1-kvark uchun S va S' sistemalaridagi burilish nuqtalarining koordinatalari quyidagi jadvalda keltirilgan:

S sistema		S' sistema	
x_1	t_1	$x'_1 = \gamma(x_1 + \beta ct_1)$ $= \frac{5}{4}x_1 + \frac{3}{4}ct_1$	$t'_1 = \gamma(t_1 + \beta x_1/c)$ $= \frac{5}{4}t_1 + \frac{3}{4}x_1/c$
0	0	0	0
L	τ	$\gamma(1 + \beta)L = 2L$	$\gamma(1 + \beta)\tau = 2\tau$
0	2τ	$2\gamma\beta L = \frac{3}{2}L$	$2\gamma\tau = \frac{5}{2}\tau$
$-L$	3τ	$\gamma(3\beta - 1)L = L$	$\gamma(3 - \beta)\tau = 3\tau$
0	4τ	$4\gamma\beta L = 3L$	$4\gamma\tau = 5\tau$

bu yerda $L = p_0 c / f = Mc^2 / (2f)$, $\tau = p_0 / f = Mc / (2f)$.

2-kvark uchun:

S sistema		S' sistema	
x_2	t_2	$x'_2 = \gamma(x_2 + \beta ct_2)$ $= \frac{5}{4}x_2 + \frac{3}{4}ct_2$	$t'_2 = \gamma(t_2 + \beta x_2/c)$ $= \frac{5}{4}t_2 + \frac{3}{4}x_2/c$
0	0	0	0
$-L$	τ	$-\gamma(1 - \beta)L = -\frac{1}{2}L$	$\gamma(1 - \beta)\tau = \frac{1}{2}\tau$
0	2τ	$2\gamma\beta L = \frac{3}{2}L$	$2\gamma\tau = \frac{5}{2}\tau$
L	3τ	$\gamma(3\beta + 1)L = \frac{7}{2}L$	$\gamma(3 + \beta)\tau = \frac{9}{2}\tau$
0	4τ	$4\gamma\beta L = 3L$	$4\gamma\tau = 5\tau$

Yuqoridagi natijalarga asosan, ikki kvarkning (x', t') diagrammalari 5-rasmda korsatilgan.

OA va OB togri chiziqlarining tenglamalari:

$$x'_1(t') = ct'; \quad 0 \leq t' \leq \gamma(1 + \beta)\tau = 2\tau; \quad (14a)$$

$$x'_2(t') = -ct'; \quad 0 \leq t' \leq \gamma(1 - \beta)\tau = \frac{1}{2}\tau. \quad (14b)$$

Ikki kvark orasidagi masofa maksimal qiymatiga $t' = \frac{1}{2}\tau$ da erishadi, shuning uchun maksimal masofa:

$$d' = 2c\gamma(1 - \beta)\tau = 2\gamma(1 - \beta)L = \frac{Mc^2}{2f}. \quad (15)$$

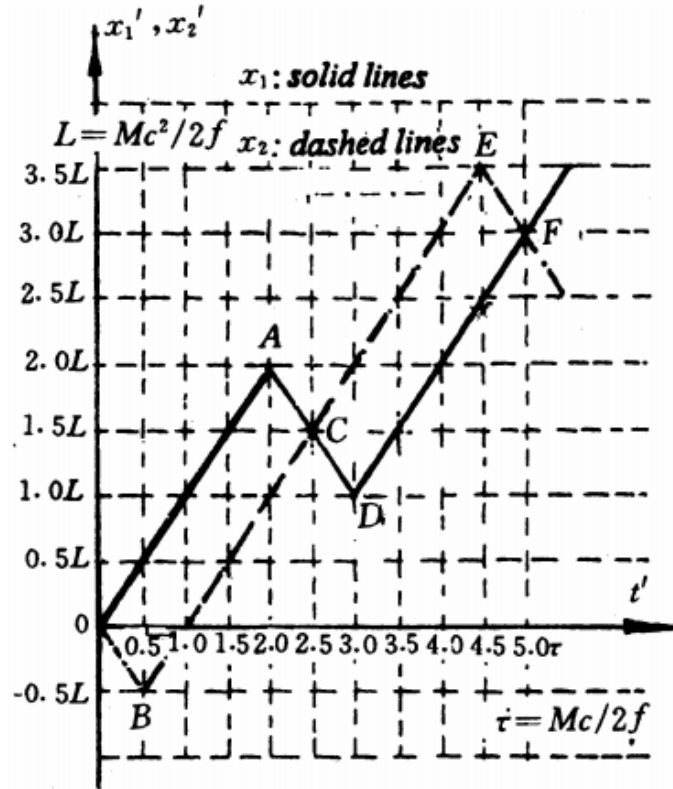


Figure 13: 5-rasm.

4)

Mezon laboratoriya sistemasiga nisbatan $V = 0.6c$ tezlik bilan harakatlanadi. Uning laboratoriya sistemasidagi energiyasi:

$$E' = \frac{Mc^2}{\sqrt{1 - \beta^2}} = \frac{1}{0.8} \times 140 = 175 \text{ MeV}.$$

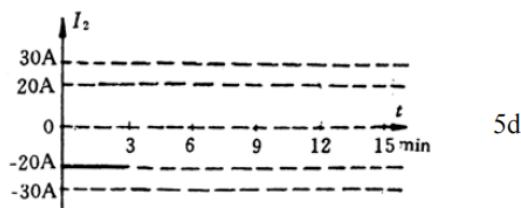
Baholash sxemasi

1-qism 2 ball, quyidagicha taqsimlangan:

- 0.4 ball – 1-rasmdagi $x(t)$ egri chizigining shakli uchun;
- 0.3 ball – 1-rasmda 4 ta teng interval uchun;
- (faqat formulani to'g'ri keltirish uchun 0.3 ball)
- 0.1 ball – A va C burilish nuqtalarining har bir koordinatasi uchun, jami 0.4 ball;
- 0.4 ball – 2-rasmdagi $p(x)$ egri chizigining shakli uchun; (faqat to'g'ri keltirish uchun 0.2 ball)
- 0.1 ball – p_0 , $L = p_0c/f$, $-p_0$, $-L$ va strelkalarning har bir to'g'ri korsatilishi uchun, jami 0.5 ball.
- (faqat burilish nuqtalari koordinatalarini to'g'ri hisoblash uchun har biri 0.05 ball).

2-qism 4 ball, quyidagicha taqsimlangan:

- 0.6 ball – $x_1(t)$ va $x_2(t)$ egri chiziqlarining har biri shakli uchun, jami 1.2 ball;
- 0.1 ball – 3-rasmdagi A , B , D va E burilish nuqtalarining har bir koordinatasi uchun, jami 0.8 ball;



Nazariy masala 2 – Yechim

1. $t = t_1$ dan t_3 gacha $r = 0$ bo'lgani uchun, magnitdagi kuchlanish $V_M = LdI_1/dt = 0$, shuning uchun

$$I_1 = I_1(t_1) = \frac{1}{2}I_0;$$

$$I_2 = I - I_1 = I - \frac{1}{2}I_0.$$

$t = t_3$ dan t_4 gacha $t = t_3$ da $I_2 = 0$ va $I \frac{1}{2}I_0$ da saqlanadi, $V_M = I_2r_n = 0$, shuning uchun I_1 va I_2 o'zgarmaydi:

$$I_1 = \frac{1}{2}I_0; \quad I_2 = 0.$$

Natijalar 6-rasmda ko'rsatilgan.

2. $t = 0$ dan $t = 1$ min gacha: $r = 0$, $V_M = 0 \Rightarrow I_1 = I_1(0) = 0$, $I_2 = I - I_1 = 0.5$ A. $t = 1$ min da r 0 dan r_n ga sakraydi, $I E/R$ dan $E/(R+r_n)$ ga oniy pasayadi, chunki I_1 induktivlik tufayli keskin o'zgarmaydi. $E/R = 0.5$ A, $R = 7.5 \Omega$, $r_n = 5 \Omega$ bo'lsa, I 0.3 A ga tushadi. $t = 1$ min dan 2 min gacha: I , I_1 , I_2 asta-sekin o'zlarining barqaror qiymatlariga yaqinlashadi:

$$I = E/R = 0.5 \text{ A}, \quad I_1 = I = 0.5 \text{ A}, \quad I_2 = 0.$$

Vaqt doimiysi $\tau = L(R + r_n)/(Rr_n) = 10 \cdot 12.5/(37.5) = 3.33$ s (taxminan 3 s). $t = 2$ min dan 3 min gacha: $r = 0$ bo'lgani uchun I_1 va I_2 o'zgarmaydi: $I_1 = 0.5$ A, $I_2 = 0$.

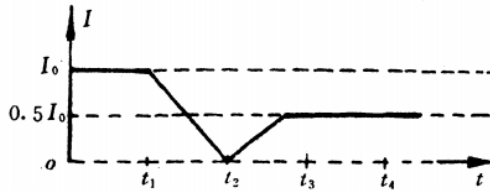
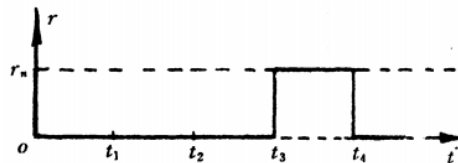
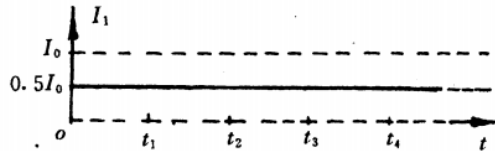


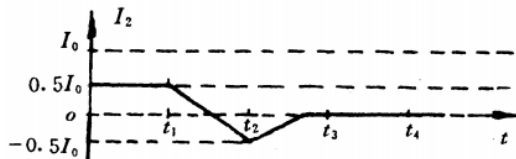
Fig. 6a



6b



6c



6d

6a-6d rasmlar

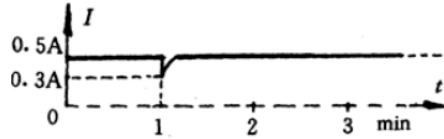
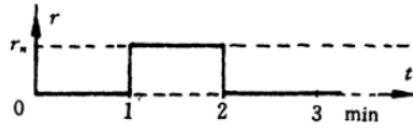
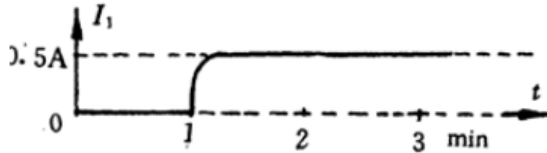


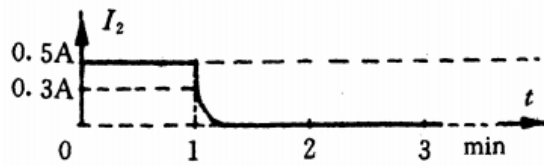
Fig. 7a



7b



7c



7d

7a-7d rasmlar

Amallar ketma-ketligi: 1. Quvvat kaliti K ni yoqing va umumiy tok I ni 20 A ga (ya'ni I_1 ga teng) oshiring. $r = 0$ bo'lgani uchun $V_M = 0$, I_1 o'zgarmaydi, I_2 20 A ga oshadi (ya'ni I_2 -20 A dan 0 ga o'zgaradi). 2. r ni 0 dan r_n ga o'tkazing. 3. $I_2 < 0.5$ A shartini saqlagan holda I ni asta-sekin nolga kamaytiring: $I_2 = V_M/r_n$, $V_M = LdI_1/dt$, $L = 10$ H, $r_n = 5 \Omega$ bo'lsa, $I_2 < 0.5$ A sharti $dI_1/dt < 0.25$ A/s ni talab qiladi, ya'ni 1 daqiqada 15 A dan kam o'zgarish. 4. $V_M = 0$ bo'lganda r ni nolga o'tkazing va K ni o'chiring. Natijalar 8-rasmda.

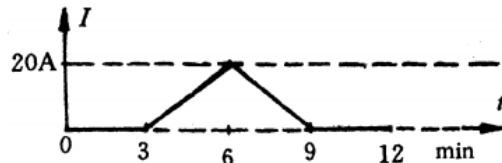
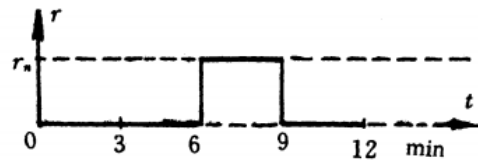
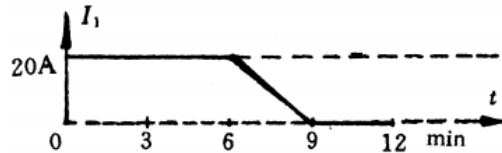


Fig. 8a



8b



8c

Figure 14: 8a-8c rasmlar

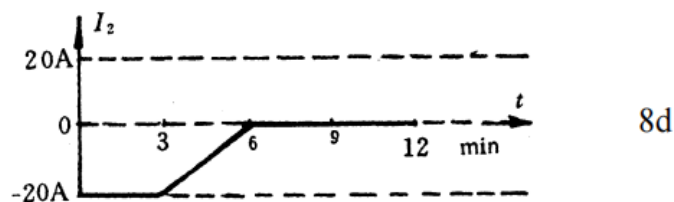


Figure 15: 8a-8c rasmlar

1 va 2-amallar 3-qismdagi kabi, $I_2 = 0$ ga erishiladi. 3. $I_2 < 0.5$ A talabi bilan I ni 10 A ga (30 A ga) oshiring. 4. $V_M = 0$ da r ni nolga o'tkazing. 5. I ni nolga kamaytiring, $I_1 = 30$ A o'zgarmaydi ($V_M = 0$), $I_2 = I - I_1 - 30$ A ga o'zgaradi. Magnitdagi tok supero'tkazgich kalit orqali yopiladi. 6. Quvvat kaliti K ni o'chiring. Magnit persistent rejimda ishlaydi. Natijalar 9-rasmda.

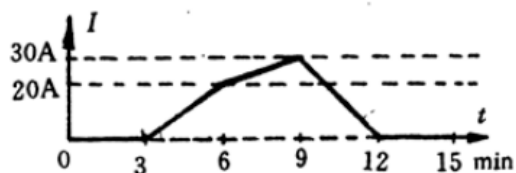
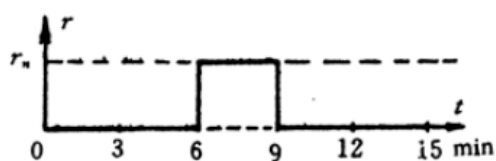
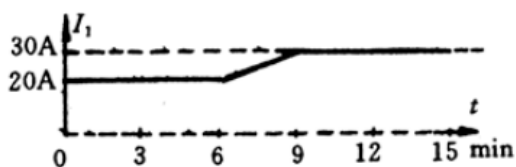


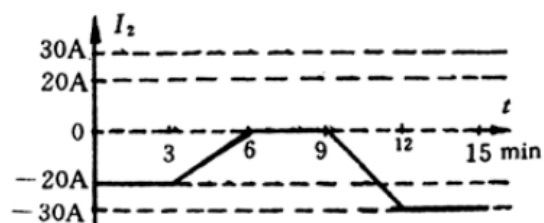
Fig. 9a



9b



9c



9d

Figure 16: 9a-9c rasmlar

Baholash sxemasi

1-qism, 2 ball:

- $t = t_1$ dan t_3 gacha bolgan I_1 va I_2 ning har biri uchun 0.5 ball;
- $t = t_3$ dan t_4 gacha bolgan I_1 va I_2 ning har biri uchun 0.5 ball.

2-qism, 3 ball:

- $t = 0$ dan 1 min gacha bolgan I_1 , I_2 ning har biri uchun 0.3 ball;

- $t = 1$ min dagi I_1, I_2, I_3 ning har biri uchun 0.3 ball;
- $t = 1$ dan 2 min gacha bolgan I_0, I_1, I_2 ning har biri uchun 0.3 ball;
- $t = 2$ dan 3 min gacha bolgan I, I_1, I_2 ning har biri uchun 0.2 ball.

3-qism, 2 ball:

- 8-rasmdagi $t = 3$ dan 9 min gacha bolgan har bir bolim uchun 0.25 ball (jami 8 bolim).

4-qism, 3 ball:

- 9-rasmdagi $t = 3$ dan 12 min gacha bolgan har bir bolim uchun 0.25 ball (jami 12 bolim).

Nazariy masala 3 – Yechim

1. A diski B diski bilan to'qnashganda, n – tutashish nuqtasidagi sirtlarga normal birlik vektor, t – tangensial birlik vektor bo'lsin. φ – n va x o'qi orasidagi burchak. U holda

$$b = (R_A + R_B) \sin \varphi.$$

To'qnashuvdan oldin A va B ning n va t bo'ylab impuls komponentlari:

$$mV_{An} = mV \cos \varphi, \quad mV_{Bn} = 0, \quad mV_{At} = mV \sin \varphi, \quad mV_{Bt} = 0.$$

To'qnashuvdan keyingi komponentlarni mV'_{An} , mV'_{Bn} , mV'_{At} , mV'_{Bt} deb belgilaymiz. ω_A , ω_B – to'qnashuvdan keyingi burchak tezliklar, $I_A = \frac{1}{2}mR_A^2$, $I_B = \frac{1}{2}mR_B^2$. Impuls saqlanishi:

$$mV \cos \varphi = mV'_{An} + mV'_{Bn}, \quad (1)$$

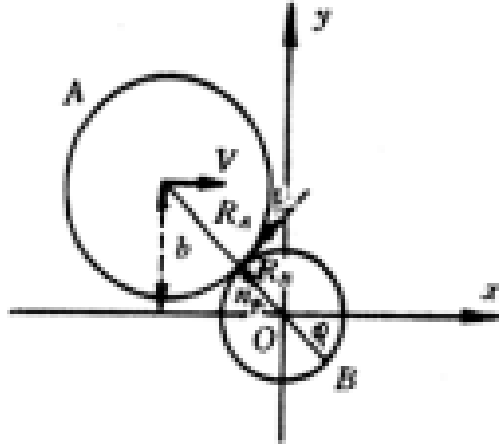
$$mV \sin \varphi = mV'_{At} + mV'_{Bt}. \quad (2)$$

O nuqtasiga nisbatan burchak impulsining saqlanishi:

$$mVb = mV'_{At}(R_A + R_B) + I_A\omega_A + I_B\omega_B. \quad (3)$$

Ishqalanish kuchi impulsi B ga t yo'nalishida impuls o'zgarishini keltirib chiqaradi va bir vaqtda $I_B\omega_B$ hosil qiladi:

$$mV'_{Bt}R_B = I_B\omega_B. \quad (4)$$



To'qnashuv paytida tutashish nuqtasida A va B bir xil tangensial tezlikka ega bo'ladi:

$$V'_{At} - \omega_A R_A = V'_{Bt} - \omega_B R_B. \quad (5)$$

Normal yo'nalishdagi nisbiy tezliklarning kattaliklari to'qnashuvdan oldin va keyin teng:

$$V \cos \varphi = V'_{Bn} - V'_{An}. \quad (6)$$

(1) va (6) dan:

$$V'_{An} = 0, \quad V'_{Bn} = V \cos \varphi.$$

(2)-(5) dan:

$$V'_{At} = \frac{5}{6}V \sin \varphi, \quad V'_{Bt} = \frac{1}{6}V \sin \varphi,$$

$$\omega_A = \frac{V \sin \varphi}{3R_A}, \quad \omega_B = \frac{V \sin \varphi}{3R_B}.$$

To'qnashuvdan keyingi tezliklarning x va y komponentlari:

$$V'_{Ax} = V'_{An} \cos \varphi + V'_{At} \sin \varphi = \frac{5Vb^2}{6(R_A + R_B)^2}, \quad (7)$$

$$V'_{Ay} = -V'_{An} \sin \varphi + V'_{At} \cos \varphi = \frac{5Vb\sqrt{(R_A + R_B)^2 - b^2}}{6(R_A + R_B)^2}, \quad (8)$$

$$V'_{Bx} = V'_{Bn} \cos \varphi + V'_{Bt} \sin \varphi = V \left[1 - \frac{5b^2}{6(R_A + R_B)^2} \right], \quad (9)$$

$$V'_{By} = -V'_{Bn} \sin \varphi + V'_{Bt} \cos \varphi = -\frac{5Vb\sqrt{(R_A + R_B)^2 - b^2}}{6(R_A + R_B)^2}. \quad (10)$$

To'qnashuvdan keyingi kinetik energiyalar:

$$E'_A = \frac{1}{2}m(V'_{Ax}{}^2 + V'_{Ay}{}^2) + \frac{1}{2}I_A\omega_A^2 = \frac{3mV^2b^2}{8(R_A + R_B)^2}, \quad (11)$$

$$E'_B = \frac{1}{2}m(V'_{Bx}{}^2 + V'_{By}{}^2) + \frac{1}{2}I_B\omega_B^2 = \frac{1}{2}mV^2 \left[1 - \frac{11b^2}{12(R_A + R_B)^2} \right]. \quad (12)$$

Baholash sxemasi

1. To'qnashuvdan keyin A va B disklarining tezlik komponentlari yechimning 7, 8, 9 va 10-tenglamalarida ko'rsatilgan. Ushbu qismning umumiy balli 8.0 ga teng. Agar to'rtta tezlik komponenti ham to'g'ri bo'lgan natija olinmagan bo'lsa, ball quyidagi qoidalarga asosan beriladi:

- Har bir to'g'ri tezlik komponenti uchun 0.8 ball;
- Disklarning markazlarini tutashtiruvchi chiziq bo'ylab nisbiy tezliklarining kattaliklari to'qnashuvdan oldin va keyin bir xil ekanligining to'g'ri ta'rifi uchun 0.8 ball;
- Burchak impulsining saqlanishining to'g'ri ta'rifi uchun 0.8 ball;
- Tegish nuqtasidagi tangensial tezliklarning tengligining to'g'ri ta'rifi uchun 0.8 ball;
- Impuls va impuls momenti o'rtasidagi munosabatning to'g'ri ta'rifi uchun 0.8 ball.

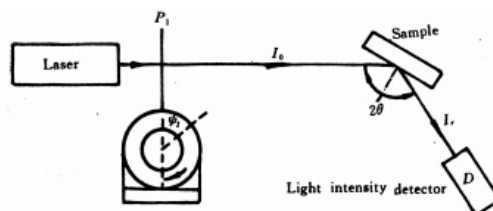
2. To'qnashuvdan keyin A disk va B diskning kinetik energiyalari yechimning 11 va 12-tenglamalarida ko'rsatilgan.

- A diskning to'g'ri kinetik energiyasi uchun 1.0 ball;
- B diskning to'g'ri kinetik energiyasi uchun 1.0 ball.

Ushbu qismning umumiy balli 2.0 ga teng.

Amaliy masala 1 – Yechim

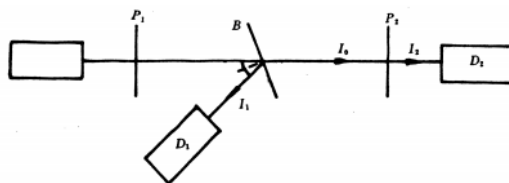
- (a) Qutblagichning o'tkazish o'qini va namunaning Brewster burchagi θ_B ni $R_p = 0$ bo'lgan Brewster burchagida p -komponentaning qaytarish koeffitsienti nolga teng ekanligidan foydalanib aniqlang. P_1 ning o'tkazish o'qi yo'nalishini (darajali shkala diskidagi belgilangan chiziq holati ψ) va tushish burchagi θ_i ni $I_r = 0$ bo'lguncha ketma-ket o'zgartiring.



Endi tushayotgan yorug'lik faqat p -komponentadan iborat va tushish burchagi θ_B ga teng. Tegishli ψ_1 va θ_B qiymatlari:

ψ_1	140.0°	322.0°	141.0°	322.5°
θ	56.4°	56.4°	56.2°	56.2°

$\psi_1 = 140.5^\circ \pm 0.5^\circ$ yoki $322.3^\circ \pm 0.1^\circ$. Brewster burchagi $\theta_B = 56.3^\circ \pm 0.1^\circ$.



Yorug'lik intensivligi va mikroampermetr ko'rsatkichi o'rtasidagi chiziqli bog'liqlikni tekshirish. Ikki qutblagich P_1 va P_2 dan o'tgan yorug'lik intensivligi Malus qonuniga bo'ysunadi:

$$I = I_0 \cos^2 \theta,$$

bu yerda I_0 – P_1 tomonidan qutblangan va tushayotgan yorug'lik intensivligi, I – o'tgan yorug'lik intensivligi, θ – P_1 va P_2 o'tkazish o'qlari orasidagi burchak. Shunday qilib, ikki qutblagich yordamida tekshirish uchun turli intensivlikdagi yorug'lik olish mumkin. Tajriba sxemasi rasmda ko'rsatilgan. D_1 tushayotgan nurning intensivlik tebranishini kuzatadi (I_1/I_2 nisbati o'zgarmaydi), D_2 esa I_2 ni o'lchaydi. $i_1(\theta)$ va $i_2(\theta)$ – mos ravishda D_1 va D_2 ko'rsatkichlari, $\psi_2(\theta)$ – belgilangan chiziq holati. $\theta = 90^\circ$ da $i_2 = 0$, mos $\psi_2(90^\circ) = 4^\circ$.

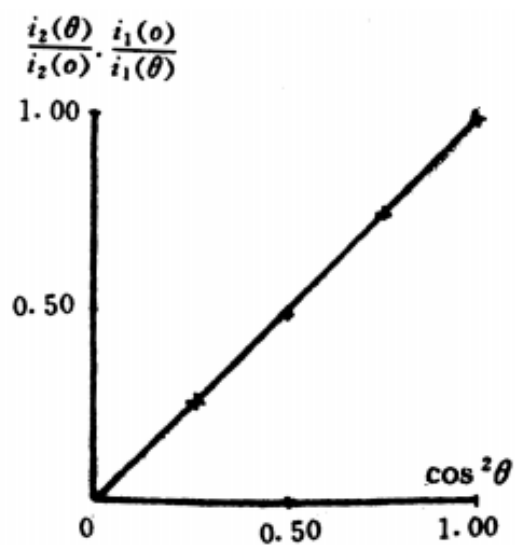
$$\theta = |\psi_2 - \psi_2(90^\circ) \pm 90^\circ|.$$

Ma'lumotlar:

ψ_2	94.0°	64.0°	49.0°	34.0°	4.0°
θ	0.0°	30.0°	45.0°	60.0°	90.0°
$i_1(\theta) \mu A$	6.3×1	5.7×1	5.7×1	5.7×1	5.7×1
$i_2(\theta) \mu A$	18.7×5	12.7×5	8.2×5	4.0×5	0.0×5

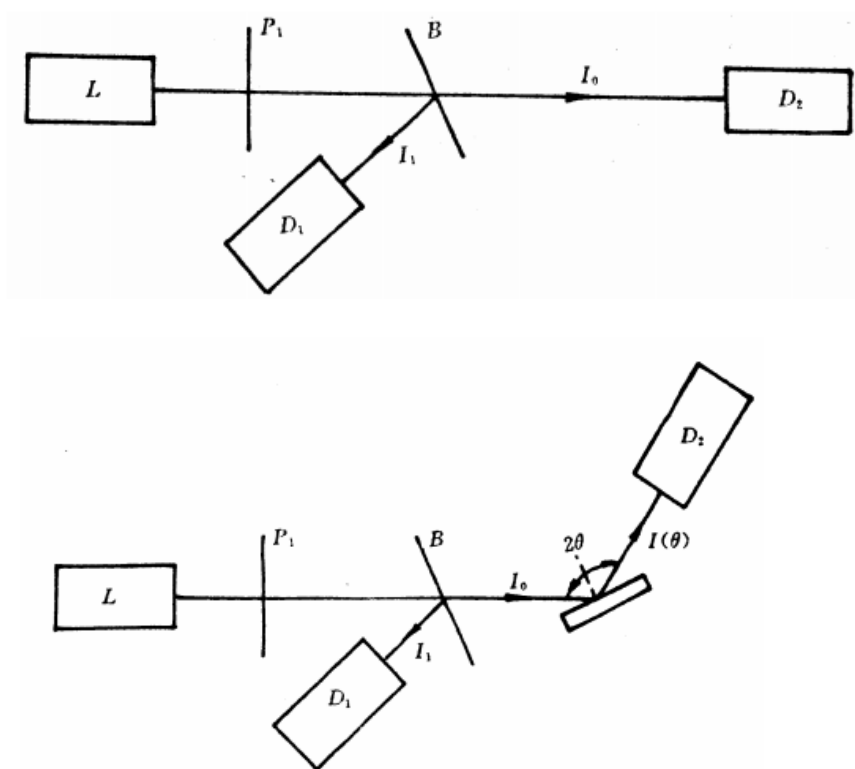
$I(\theta)/I_0 = \frac{i_2(\theta)}{i_1(\theta)} \cdot \frac{i_1(0)}{i_2(0)}$ dan hisoblanadi:

θ	0°	30°	45°	60°	90°
$\cos^2 \theta$	1.00	0.75	0.50	0.25	0.00
$I(\theta)/I_0$	1.00	0.75	0.49	0.24	0.00



Bu chiziqli bog'liqlikni tasdiqlaydi.

Qaytarish koeffitsientini o'lchash. I_0 va I_1 nisbatini aniqlash uchun sxema qo'llaniladi: $I_0/I_1 \propto i_{20}/i_{10}$. Keyin p -komponenta bilan turli tushish burchaklarida R_p o'lchanadi. $i_1(\theta)$, $i_2(\theta)$ – D_1 va D_2 ko'rsatkichlari.



$$R_p(\theta) = \frac{I(\theta)}{I_0} = \frac{i_2(\theta)}{i_1(\theta)} \cdot \frac{i_{10}}{i_{20}}.$$

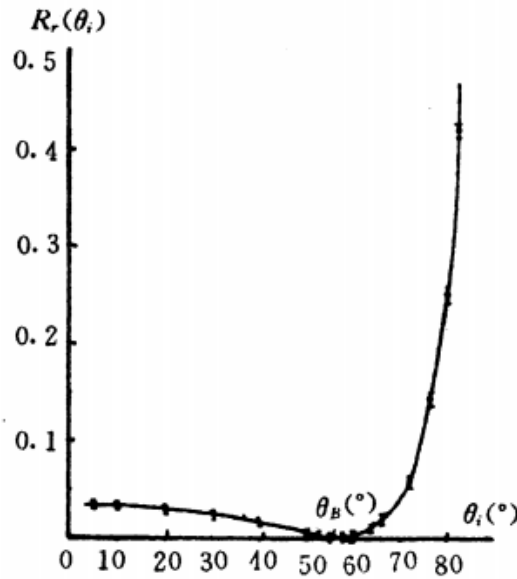
$\psi_1 = 140.5^\circ$, $i_{20} = 19.8 \times 5 \mu\text{A}$, $i_{10} = 13.3 \mu\text{A}$. Jadval:

$\theta(^{\circ})$	$i_2(\theta)$	$i_1(\mu\text{A})$	$R_p(\theta)$
5	15.1×0.2	11.1	0.037
10	14.9×0.2	11.2	0.036
20	13.3×0.2	11.1	0.032
30	11.4×0.2	12.2	0.025
40	7.8×0.2	14.7	0.014
50	2.3×0.2	16.9	0.0037
53	0.7×0.2	11.3	0.0017
55	0.3×0.2	11.3	0.00059
56.3	~ 0	11.5	~ 0
58	0.3×0.2	11.5	0.0007
60	1.1×0.2	13.5	0.0024
64	6.5×0.2	16.7	0.011
66	7.8×0.2	11.8	0.018
68	16.3×0.2	15.0	0.029
72	5.3×0.1	11.7	0.061
76	13.1×1	14.0	0.13
80	4.4×5	11.7	0.25
84	9.1×5	14.5	0.42

Grafik chiziladi (pleksiglas uchun p -komponenta qaytarish koeffitsientining tushish burchagiga bog'liqligi).

Brewster burchagi yuqoridagi ma'lumotlardan $\theta_B = 56.3^\circ \pm 0.2^\circ$. Sindirish ko'rsatkichi:

$$n = \tan \theta_B = 1.50 \pm 0.01.$$



Xato manbalari: detektor sezgirligi past; tushayotgan yorug'lik faqat p -komponentadan iborat emas; daraja shkalalari bir xil emas.

Amaliy masala 2 – Yechim

1. Zanjir diagrammasi 1-rasmda ko'rsatilgan.

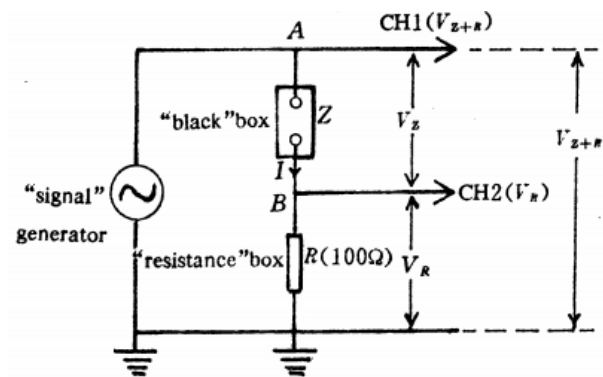
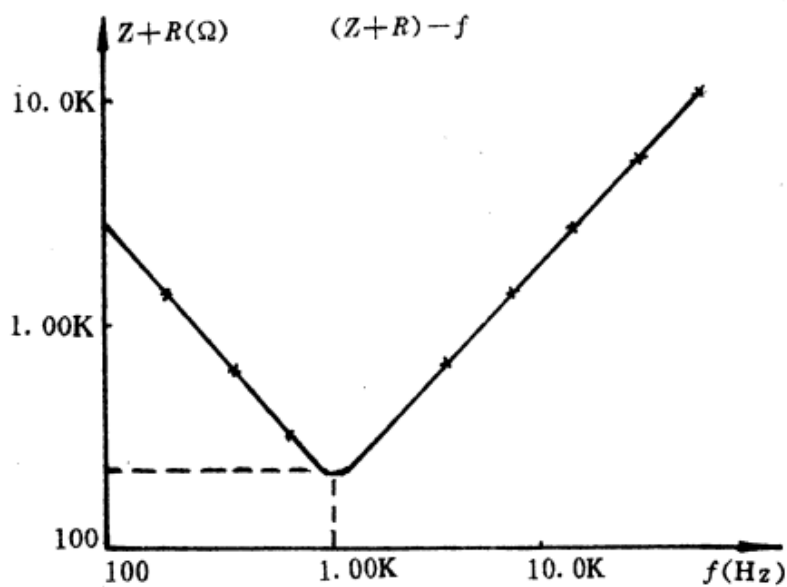


Fig. 1

1-rasm: O'lchash zanjiri

$$I = \frac{V_R}{R}, \quad Z + R = \frac{V_{Z+R}}{I} = \frac{V_{Z+R}}{V_R} R.$$



2-rasm: $Z + R$

chastota funksiyasi

Turli chastotalarda V_{Z+R} va V_R ni o'lchang. O'lchangan ma'lumotlar va hisoblangan $Z+R$ jadvali:

$f \times 10^3 \text{ Hz}$	$U_{Z+R} (V_{pp})$	$U_R (\text{mV}_{pp})$	$Z + R \times 10^3 \Omega$
0.100	0.600	22.0	2.73
0.200	0.600	45.0	1.33
0.400	0.600	94.0	0.638
0.700	0.300	92.0	0.326
0.900	0.300	121	0.248
1.00	0.300	136	0.220
1.10	0.300	140	0.214
1.16	0.300	141	0.213
1.25	0.300	140	0.214
1.50	0.300	120	0.250
2.00	0.300	88.0	0.341
4.00	0.300	78.0	0.769
8.00	0.600	38.0	1.58
15.0	0.600	20.0	3.00
30.0	0.600	10.0	6.00
50.0	0.600	6.0	10.0

$Z + R \propto f$ egri chizig'i 2-rasmda chizilgan. Xulosa: (1) Tok rezonansi (Z minimal) $f_0 \approx 1.16 \times 10^3 \text{ Hz}$ da. (2) $f < f_0$ da $Z \propto f$, $\Delta\varphi \approx -\pi/2$ – past chastotalarda impedans induktivlik bilan aniqlanadi. (3) $f > f_0$ da $Z \propto f$, $\Delta\varphi \approx \pi/2$ – yuqori chastotalarda impedans sig'im bilan aniqlanadi. (4) Qora qutining ekvivalent zanjiri – ketma-ket ulangan r , L , C (3-rasm).

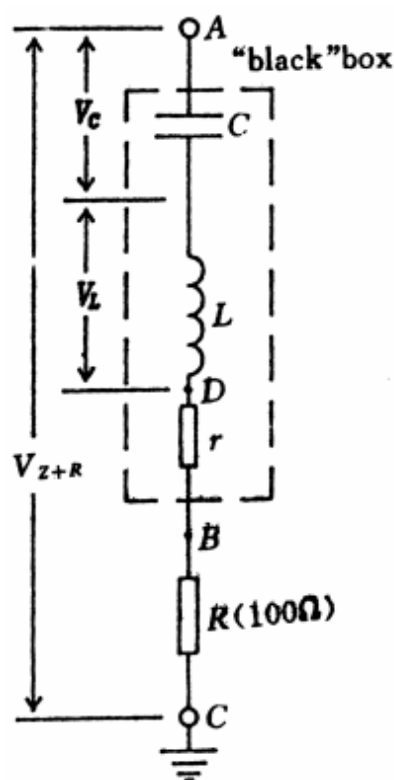


Fig. 3

Figure 17: 3-rasm: Ekvivalent zanjir

Qiymatlarni aniqlash:

- (a) r : Rezonans chastotasida $Z + R = r + R$. Jadvaldan $r + R = 213 \Omega$, $R = 100 \Omega$ berilgan, shuning uchun $r = 113 \Omega$.
- (b) C : Past chastotalarda $z_L \approx 0$, zanjir ketma-ket RC . $1/(\omega C) = Z_C = V_C/I \approx V_{Z+R}/I = Z + R$. $f = 100 \text{ Hz}$ da $Z + R = 2.73 \times 10^3 \Omega$:

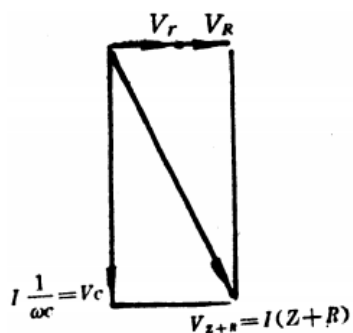


Figure 18: 4-rasm: Past chastotadagi fazor diagrammasi

$$C \approx \frac{1}{\omega(Z+R)} = \frac{1}{2\pi \cdot 100 \cdot 2.73 \times 10^3} \approx 0.58 \mu\text{F}.$$

- (c) L : Yuqori chastotalarda $Z_C \approx 0$, zanjir ketma-ket RL . $\omega L = Z_L = V_L/I \approx V_{Z+R}/I = Z+R$. $f = 50 \text{ kHz}$ da $Z+R = 10^4 \Omega$:

$$L = \frac{Z+R}{\omega} = \frac{10^4}{2\pi \cdot 50 \times 10^3} \approx 31.8 \text{ mH}.$$

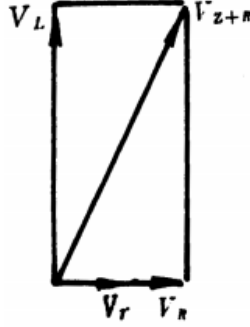


Fig. 5

Figure 19: 5-rasm: Yuqori chastotadagi fazor diagrammasi

Xato baholash

Berilgan:

$$\frac{\Delta R}{R} \approx 0.5\%, \quad \frac{\Delta V}{V} \approx 4\%.$$

1. Qarshilik r : rezonans chastotasi f_0 da

$$r+R = \frac{V_{Z+R}}{V_R} R,$$

$$\frac{\Delta(r+R)}{r+R} = \frac{\Delta V_{Z+R}}{V_{Z+R}} + \frac{\Delta V_R}{V_R} + \frac{\Delta R}{R} \approx 4\% + 4\% + 0.5\% = 8.5\%,$$

$$\Delta r = 16 \Omega.$$

2. Sig'im C (chastota o'qish xatosini hisobga olmang):

$$\frac{1}{\omega C} \approx Z_C = \frac{V_{Z+R}}{V_R} R,$$

$$\frac{\Delta C}{C} = \frac{\Delta V_{Z+R}}{V_{Z+R}} + \frac{\Delta V_R}{V_R} + \frac{\Delta R}{R} \approx 8.8\%.$$

$V_C \approx V_{Z+R}$ yaqinlashishi 0.3% foizli xato kiritadi.

3. Induktivlik L : sig'imdagi natijalarga o'xshash, ammo $V_L \approx V_{Z+R}$ yaqinlashishi kiritgan foizli xato juda kichik (0.003%) va shuning uchun ahamiyatsiz.

$$\frac{\Delta L}{L} \approx 8.5\%.$$

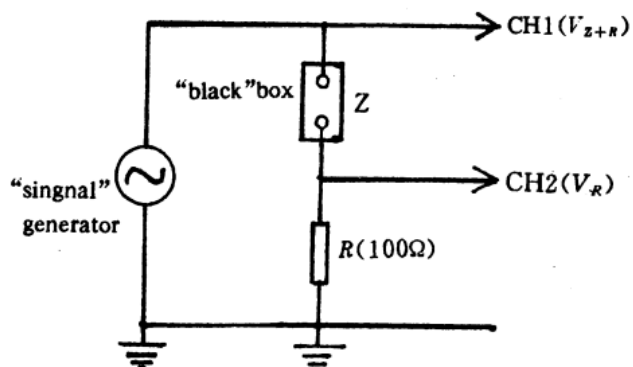


Fig. a

Eksperimental masala 2: Baholash sxemasi (maksimal 10 ball)

1. O'lchash zanjiri (a)-rasmda ko'rsatilganidek to'g'ri 2.0 ball
2. Qora qutining xarakteristikasini ko'rsatadigan to'g'ri ma'lumotlar jadvali va grafik 2.0 ball
3. Qora qutining ekvivalent sxemasi va undagi elementlarning nomlari va qiymatlari to'g'ri (jami 6.0 ball)
 - (a) R, L va C ketma-ket ulangan 1.5 ball
(L va C ketma-ket ulangan 1.0 ball)
 - (b) Har bir element uchun to'g'ri qiymat (xato 15% dan kam) 0.5 ball ($\times 3$)
(xato 15% va 30% orasida 0.3 ball; xato 30% va 50% orasida 0.1 ball)
 - (c) Har bir element uchun to'g'ri hisoblash formulasi 0.5 ball ($\times 3$)
 - (d) Har bir element uchun xato bahosi asosli 0.5 ball ($\times 3$)